

CONTACT INFORMATION	Biomedical Mathematics Group Institute for Basic Science (IBS) 55 Expo-ro Yuseong-gu Daejeon 34126, South Korea	<i>E-mail:</i> hyukpyo.hong13@gmail.com <i>Web:</i> https://hyukpyohong.github.io
APPOINTMENTS	Institute for Basic Science (IBS) , Daejeon, South Korea Visiting Research Fellow, Biomedical Mathematics Group May 2026–present University of Wisconsin–Madison , Madison, Wisconsin, USA Van Vleck Assistant Professor, Department of Mathematics Aug. 2023–May 2026	
EDUCATION	KAIST , Daejeon, South Korea Ph.D. in Mathematical Sciences Feb. 2018–Aug. 2023 Advisor: Jae Kyoung Kim Thesis: Development of stochastic model reduction framework for analysis and inference of biochemical reaction networks B.S. in Mathematical Sciences Mar. 2013–Feb. 2018	
RESEARCH INTERESTS	Fields: Mathematical biology, ODEs, Stochastic processes, Bayesian statistics, Digital medicine Topics: Steady states of ODEs, stationary distributions of CTMCs, Koopman operator theory, MCMC methods, parameter estimation for non-Markovian stochastic models, metabolic control analysis, homeostasis and adaptation in biological systems, neurodegenerative disease-related alteration of human motor activity	
PAPERS	†: (co-)1st author, *: (co-)corresponding author. Note: Where no author is designated as the first author (†), names are listed in alphabetical order by last name, as is standard practice in mathematical journals. In preparation or preprint: 1. Ella Buelling†, Wolfram Liebermeister*, Hyukpyo Hong*, Connections between structural sensitivity analysis and metabolic control analysis, <i>in preparation</i> 2. Hyukpyo Hong, Seewoo Lee, Necessary and sufficient conditions of the existence of a finite dimensional invariant subspace for polynomial ODEs, <i>in preparation</i> 3. Geonhee Ryu†, Hyukpyo Hong*, Daewook Kim*, Non-autonomous dynamics learning via Koopman-based Neural Networks, <i>in preparation</i> 4. Hyukpyo Hong†, Qin Li†, Matthew Colbrook, Hanbaek Lyu*, Finding Koopman Invariant Subspaces via Personalized PageRank, <i>Submitted</i> 5. Hyukpyo Hong†*, Diego Rojas La Luz†, Gheorghe Craciun*, Ubiquitous asymptotic robustness in biochemical systems, <i>Under review</i>; [arXiv] Published or accepted: 1. Dongju Lim†, Kyeong Tae Ko†, Hyukpyo Hong, Hyojung Lee, Boseung Choi, Won Chang, Sunhwa Choi*, Jae Kyoung Kim*, A history-dependent approach for accurate initial condition estimation in epidemic models, <i>PLoS Computational Biology</i>, 2025; [journal] 2. Hyukpyo Hong†, Seokhwan Moon†, Yuji Hirono†*, Jae Kyoung Kim*, Topological criterion for robust perfect adaptation of reaction fluxes in biological networks, <i>iScience</i>, 2025; [arXiv]; [journal]	

3. **Hyukpyo Hong**[†], Eunjin Eom[†], Hyojung Lee, Sunhwa Choi*, Boseung Choi*, Jae Kyoung Kim*, Overcoming bias in estimating epidemiological parameters with realistic history-dependent disease spread dynamics, *Nature Communications*, 2024; [\[journal\]](#)
4. Hyeontae Jo[†], **Hyukpyo Hong**[†], Hyung Ju Hwang, Won Chang, Jae Kyoung Kim*, Density physics-informed neural networks reveal sources of cell heterogeneity in signal transduction, *Patterns*, 2023; [\[bioRxiv\]](#); [\[journal\]](#)
5. **Hyukpyo Hong**[†], Mark Jayson Cortez, Yu-Yu Cheng, Hang J. Kim, Boseung Choi*, Krešimir Josić*, Jae Kyoung Kim*, Inferring delays in partially observed gene regulation processes, *Bioinformatics*, 2023; [\[bioRxiv\]](#); [\[journal\]](#)
6. **Hyukpyo Hong**, Bryan S. Hernandez, Jinsu Kim, Jae Kyoung Kim*, Computational translation framework identifies biochemical reaction networks with special topologies and their long-term dynamics, *SIAM Journal on Applied Mathematics*, 2023; [\[arXiv\]](#); [\[journal\]](#)
7. **Hyukpyo Hong**[†], Ji Yun Noh[†], Hyojung Lee, Sunhwa Choi, Boseung Choi, Jae Kyoung Kim*, Eui-Cheol Shin*, Modeling incorporating the severity-reducing long-term immunity: higher viral transmission paradoxically reduces severe COVID-19 during endemic transition, *Immune Network*, 2022; [\[medRxiv\]](#); [\[journal\]](#)
8. Dae Wook Kim[†], **Hyukpyo Hong**[†], Jae Kyoung Kim*, Systematic inference identifies a major source of heterogeneity in cell signaling dynamics: the rate-limiting step number, *Science Advances*, 2022; [\[journal\]](#)
9. Yun Min Song[†], **Hyukpyo Hong**[†], Jae Kyoung Kim*, Universally valid reduction of multiscale stochastic biochemical systems with simple non-elementary propensities, *PLoS Computational Biology*, 2021; [\[bioRxiv\]](#); [\[journal\]](#)
10. Mark Jayson Cortez[†], **Hyukpyo Hong**, Boseung Choi*, Jae Kyoung Kim*, Krešimir Josić*, Hierarchical Bayesian models for inference in biochemical reactions with delays, *Bioinformatics*, 2021; [\[bioRxiv\]](#); [\[journal\]](#)
11. Jaehyoung Hong[†], Su Jung Choi[†], Se Ho Park, **Hyukpyo Hong**, Victoria Booth, Eun Yeon Joo*, Jae Kyoung Kim*, Personalized sleep-wake patterns aligned with circadian rhythm relieve daytime sleepiness, *iScience*, 2021; [\[bioRxiv\]](#); [\[journal\]](#)
12. **Hyukpyo Hong**[†], Jinsu Kim[†], M. Ali Al-Radhawi, Eduardo D. Sontag, Jae Kyoung Kim*, Derivation of stationary distributions of biochemical reaction networks via structure transformation, *Communications Biology*, 2021; [\[bioRxiv\]](#); [\[journal\]](#)

BOOK CHAPTERS

1. **Hyukpyo Hong**, Boseung Choi, Jae Kyoung Kim*, Beyond the Michaelis-Menten: Bayesian inference for enzyme kinetic analysis, Computational Methods for Estimating the Kinetics Parameters of Biological Systems, Methods in Molecular Biology, vol 2385. Humana, New York, NY.; [\[journal\]](#); [\[link\]](#)

RESEARCH GRANTS

2019–2023 National Research Foundation of Korea (NRF), NRF-2019-Fostering Core Leaders of the Future Basic Science Program/Global Ph.D. Fellowship Program, 2019H1A2A1075303, **Principal Investigator** (120,000 USD)
 Title: *Development of Stochastic Model Reduction Methodology and its Application to an Optogenetic System*

HONORS AND AWARDS

2025 Postdoctoral Excellence in Teaching Award, UW–Madison
 2025 Excellent Dissertation Award, Korean Mathematical Society
 2024 KSIAM Next Generation Researcher Award, Korean Society for Industrial and Applied Mathematics
 2022 SIAM Student Travel Award, Society for Industrial and Applied Mathematics
 2022 Blood Donor Hall of Fame (100 donations), Korean Red Cross

2019–2023 Global Ph.D. Fellowship (Full Tuition), National Research Foundation of Korea
 2017 36th National Undergraduate Mathematics Competition Silver Award, Korean Mathematical Society
 2016 Mirae Asset Global Exchange Scholarship, Mirae Asset Park Hyeon Joo Foundation
 2014 33rd National Undergraduate Mathematics Competition Silver Award, Korean Mathematical Society
 2014 Dean’s List Award, College of Natural Sciences, KAIST
 2013 32nd National Undergraduate Mathematics Competition Silver Award, Korean Mathematical Society
 2013–2017 National Scholarship for Science and Engineering (Full Tuition), Korea Student Aid Foundation

TEACHING

UW–Madison

- Spring 2026: [Instructor] MATH331: Introductory Probability
- Fall 2025: [Instructor] MATH331: Introductory Probability
- Spring 2025: [Instructor] MATH331: Introductory Probability
- Fall 2024: [Instructor] MATH340: Elementary Matrix and Linear Algebra
- Spring 2024: [Instructor] MATH320: Linear Algebra and Differential Equations
- Fall 2023: [Instructor] MATH320: Linear Algebra and Differential Equations

KAIST

- Spring 2019: [Teaching Assistant] MAS201: Differential Equations and Applications
- Fall 2018: [Teaching Assistant] MAS201: Differential Equations and Applications, MAS480: Introduction to Mathematical Biology
- Spring 2018: [Teaching Assistant] MAS109: Introduction to Linear Algebra, MAS212: Linear Algebra

MENTORING

UW–Madison

- Jun. 2024 – present: Ella Buelling, Junior student, Edgewood High School
Working on connections between the hypergraph representation of biochemical reaction networks and metabolic control analysis.

KAIST

- Jun. 2023 – Mar. 2025: Seokhwan Moon, Undergraduate, Department of Mathematics, POSTECH
Investigated structural conditions under which ODE steady-state solutions remain independent of system parameters. Published in *iScience*.
- Spring 2019: Minyoo Kim, Undergraduate, Department of Mathematics, KAIST
Investigated *the total quasi-steady-state approximation for a competitive system*. Won a poster prize at the 2019 KSIAM Spring Conference and a prize in the URP final evaluation.

INVITED TALKS

Apr. 28, 2026, Online, Formal reaction kinetics and related questions

Ubiquitous Asymptotic Robustness in Biochemical Systems [\[video\]](#)

Feb. 26, 2026, Online, Mathematics of Reaction Networks (MoRN) Seminar

Ubiquitous Asymptotic Robustness in Biochemical Systems [\[video\]](#)

Nov. 21, 2025, Durham, NC, USA, Duke University Math Bio Seminar

Identifying major sources of heterogeneity in antibiotic cell responses via Bayesian inference and Physics-informed neural networks

Nov. 20, 2025, Durham, NC, USA, NC State Univ. TPB Cluster
Bridging Mathematical Theory and Biological Challenges: From intracellular networks to human-scale dynamics

Nov. 19, 2025, Durham, NC, USA, NC State Univ. Dept. of Mathematics
Leveraging Koopman Operator Theory and Graph Representation for Deeper Insight into Complex Dynamical Systems

Nov. 13, 2025, Seoul, Korea, Seminar at Seoul National University
Finite-Dimensional Representations of the Koopman Operator: Equation- and Data-Driven Approaches

Oct. 31, 2025, Madison, WI, USA, UW–Madison Applied and Computational Mathematics Seminar
Finite-Dimensional Representations of the Koopman Operator: Equation- and Data-Driven Approaches

Oct. 28, 2025, Online, Midwest Mathematical Biology Seminar
Ubiquitous Asymptotic Robustness in Biochemical Systems

Oct. 1, 2025, Seoul, Korea, Seminar at Korea University
Leveraging Koopman Operator Theory and Graph Representation for Deeper Insight into Complex Dynamical Systems

Sep. 27, 2025, Chicago, IL, USA, 2025 GLSIAM Meeting
Digital biomarker of cognitive impairment: fractal patterns of human activity data

Jul. 17, 2025, Edmonton, Canada, SMB Annual Meeting
Inferring delay distributions in partially observed biological processes

Jul. 8, 2025, Madison, WI, USA, SIAM Conference on Applied Algebraic Geometry
Ubiquitous Asymptotic Robustness in Biochemical Systems

Jun. 20, 2025, Busan, Korea, Pusan National University Math Seminar
Parameter inference in Markovian and non-Markovian models

Jun. 4, 2025, Daejeon, Korea, KAIST Brain & Cognitive Sciences Seminar
Digital biomarker of cognitive impairment: fractal patterns of human activity data

Jun. 2, 2025, Seoul, Korea, Konkuk University Math Seminar
Parameter inference in Markovian and non-Markovian models

May 17, 2025, Seoul, Korea, KSIAM Spring Conference
KSIAM Next Generation Researcher Award speech

May 15, 2025, Denver, CO, USA, SIAM Conference on Applied Dynamical Systems
Network Translation of a Stochastic Reaction Network to Reveal Stationary Distributions

Mar. 3, 2025, Online, Computational and Applied Math / Data Science Seminar, Arizona State University
Koopman representation: Linear representation of nonlinear dynamics

Oct. 25, 2024, Cincinnati, OH, USA, Applied Math Seminar at the University of Cincinnati
Koopman representation: Linear representation of nonlinear dynamics

Oct. 7, 2024, Online, Seminar at the University of the Philippines Diliman
Accurate epidemiological parameters estimation with realistic history-dependent disease spread dynamics

Aug. 1, 2024, Busan, Korea, Seminar at Pusan National University
Analysis of dynamical systems using "easier" representations

Jul. 31, 2024, Daejeon, Korea, Biomedical Mathematics Seminar at Institute for Basic Science
Koopman representation: Linear representation—not an approximation—of nonlinear dynamics

Jul. 2–4, 2024, Seoul, Korea, SMB Annual Meeting

1. Inferring delays in partially observed gene regulation processes; 2. Density-PINNs reveal that multiple pathways reduce cell-to-cell heterogeneity in antibiotic responses

Jun. 4, 2024, Sejong, Korea, Data Computational Sciences Special Lecture at Korea University, Sejong Campus

How to sample random numbers from a target distribution

Jun. 3, 2024, Seoul, Korea, Data Science Seminar at Sogang University

Analysis of dynamical systems using “easier” representations

May 23, 2024, Pohang, Korea, CM2LA Seminar at POSTECH

Analysis of dynamical systems using “easier” representations

Apr. 25, 2024, Champaign, IL, USA, Mathematical Biology Seminar at UIUC

Analyzing long-term behavior of dynamical systems using “easy-to-handle” representations

Aug. 17, 2023, Daejeon, Korea, ICIAM 2023 Satellite Workshop: Stochastic Modeling and Data Analysis for Biological Systems

Network translation allows for revealing long-term dynamics of stochastic reaction networks

Jul. 17, 2023, Columbus, OH, USA, SMB Annual Meeting, Minisymposium

Network translation allows for revealing long-term dynamics of stochastic reaction networks

Jun. 27, 2023, Jeju, Korea, The 8th CIJK International Conference on Mathematical and Theoretical Biology, Minisymposium

Things we have been overlooking in infectious disease modeling

May 19, 2023, Pyeongchang, Korea, KSIAM Spring Conference, Special session

Increasing viral transmission paradoxically reduces severe COVID-19 during endemic transition

Dec. 19, 2022, Daejeon, Korea, Mini-workshop on Recent Trends in Pure and Applied Mathematics

Inferring delays in partially observed gene regulatory networks

Nov. 21, 2022, Seoul, Korea, Population Approach Group Korea Annual Meeting

Increasing viral transmission paradoxically reduces severe COVID-19 during endemic transition

Oct. 19, 2022, Seoul, Korea, Global KMS International Conference, Special session

Systematic inference identifies a major source of heterogeneity in non-Markovian cell signaling dynamics

Jun. 25, 2022, Yeosu, Korea, KSMB Annual Meeting, Special session

Increasing viral transmission paradoxically reduces severe COVID-19 during endemic transition

Jun. 22, 2022, Online, BRIC Webinar

Systematic inference for cell signalling pathways identifies a key determinant of cell-to-cell variability

May 28, 2022, Daejeon, Korea, KSIAM Spring Conference, Special session

Systematic inference for cell signalling pathways identifies a key determinant of cell-to-cell variability

Aug. 27, 2021, Online, Korea, KSMB Annual Meeting, Special session

Inference of stochastic dynamics in biochemical reaction networks

Jun. 16, 2021, Online, SMB Annual Meeting, Minisymposium

Inference of stochastic dynamics in biochemical reaction networks

May 27, 2021, Online, SIAM Conference on Applications of Dynamical Systems, Minisymposium

Derivation of stationary distributions of stochastic chemical reaction networks via network translation

May 13, 2021, Online, Seminar on the Mathematics of Reaction Networks

Derivation of stationary distributions of stochastic chemical reaction networks via network translation [\[video\]](#)

Oct. 24, 2020, Online, KMS Annual Meeting, Minisymposium

Derivation of stationary distributions of biochemical reaction networks via structure transformation

ACADEMIC
SERVICE

Jun. 2021: Support Staff, SMB Annual Meeting

Hosted Zoom sessions for the Asia time zone.

PEER REVIEW

PLoS Computational Biology, Acta Biotheoretica, Stat, PCI Math Comp Biol, Frontiers in Public Health

OUTREACH

May. 26, 2026: Alumni Career Talk, KAIST Department of Mathematical Sciences

Preparing for a Postdoc Abroad: A Three-Year Retrospective

Jun. 20, 2023: KAIST Math Graduate Student Seminar, KAIST, Daejeon, Korea

Preparation for Postdoc Positions

Sep. 16, 2022: Science Slam-D Public Lecture, Yongsan High School, Daejeon, Korea

Mathematics Meets Biology: Discovering the Optimal Sleep Pattern through Mathematical Modeling

Hosted by NST, IBS, Daejeon MBC (TV network), and Hello DD (Press). Delivered for high school students.

Apr. 27, 2021: Mobile Science Classroom, National Research Foundation of Korea

Mathematical Biology: Bridging Mathematics and the Life Sciences

Delivered for middle school students.